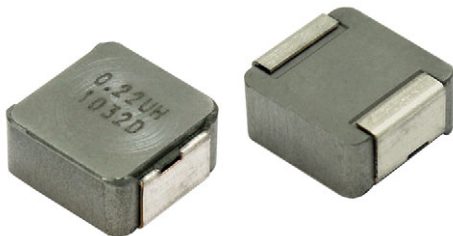


IHLP® Commercial Inductors, Low DCR Series



LINKS TO ADDITIONAL RESOURCES



RoHS
COMPLIANT
HALOGEN
FREE

FEATURES

- Shielded construction
- Excellent DC/DC energy storage up to 1 MHz to 2 MHz. Filter inductor applications up to SRF (see "Standard Electrical Specifications" table)
- Operating temperature up to 125 °C
- Lowest DCR/μH, in this package size
- Handles high transient current spikes without saturation
- Ultra low buzz noise, due to composite construction
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

APPLICATIONS

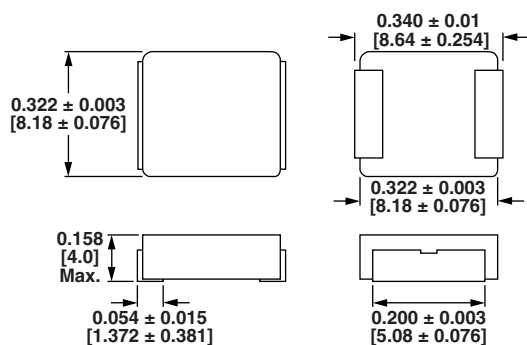
- PDA / notebook / desktop / server applications
- High current POL converters
- Low profile, high current power supplies
- Battery powered devices
- DC/DC converters in distributed power systems
- DC/DC converter for field programmable gate array (FPGA)

STANDARD ELECTRICAL SPECIFICATIONS					
L_0 INDUCTANCE ± 20 % AT 100 kHz, 0.25 V, 0 A (μH)	DCR TYP. 25 °C (mΩ)	DCR MAX. 25 °C (mΩ)	HEAT RATING CURRENT DC TYP. (A) ⁽¹⁾	SATURATION CURRENT DC TYP. (A) ⁽²⁾	SRF TYP. (MHz)
0.22	1.26	1.35	34.0	22.0	117
0.33	2.01	2.15	27.5	16.0	108
0.47	2.35	2.50	25.0	14.0	80
0.68	3.01	3.22	22.2	14.5	62
0.82	3.63	3.88	19.5	15.0	57
1.0	4.33	4.63	18.2	12.0	49
2.2	8.8	9.41	14.5	10.2	25
3.3	14.0	14.9	10.5	9.7	22
4.7	21.1	22.6	8.0	8.7	17
5.6	26.7	28.6	7.4	7.6	15
6.8	31.2	33.4	7.0	6.7	13
8.2	42.1	45.0	5.7	6.6	12.6
10.0	48.4	51.8	5.4	6.4	12
15.0	61.0	65.3	4.9	3.7	10.3
22.0	84.0	89.0	4.3	3.3	8.2
33.0	135	144	3.2	3.2	6.7

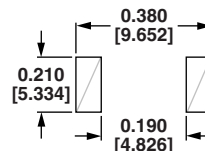
Notes

- All test data is referenced to 25 °C ambient
 - Operating temperature range -55 °C to +125 °C
 - The part temperature (ambient + temp. rise) should not exceed 125 °C under worst case operating conditions. Circuit design, component placement, PWB trace size and thickness, airflow and other cooling provisions all affect the part temperature. Part temperature should be verified in the end application.
 - Rated operating voltage (across inductor) = 50 V
- ⁽¹⁾ DC current (A) that will cause an approximate ΔT of 40 °C
⁽²⁾ DC current (A) that will cause L_0 to drop approximately 20 %

DIMENSIONS in inches [millimeters]



Typical Pad Layout





DESCRIPTION

IHLP3232DZ-11	33 μH	± 20 %	EK	e3
MODEL	INDUCTANCE VALUE	INDUCTANCE TOLERANCE	PACKAGE CODE	JEDEC® LEAD (Pb)-FREE STANDARD

GLOBAL PART NUMBER

I H L P	3 2 3 2 D Z	E K	3 3 0	M	1 1
PRODUCT FAMILY	SIZE	PACKAGE CODE	INDUCTANCE VALUE	INDUCTANCE TOLERANCE	SERIES
		EK = tape and reel	330 = 33 μ H	M = ± 20 %	

PACKAGE CODE OPTIONS

EK = tape and reel packaging (1250 pcs on 13-inch reel)

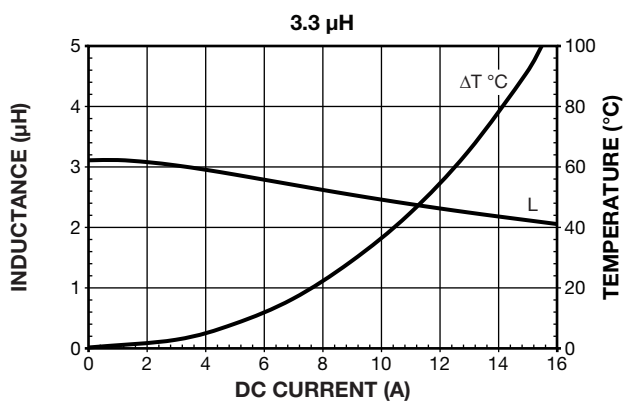
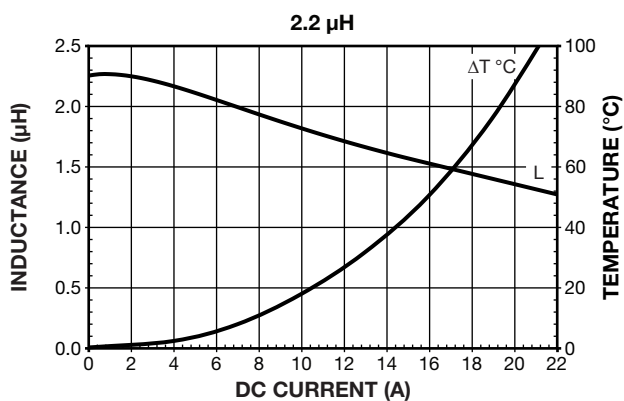
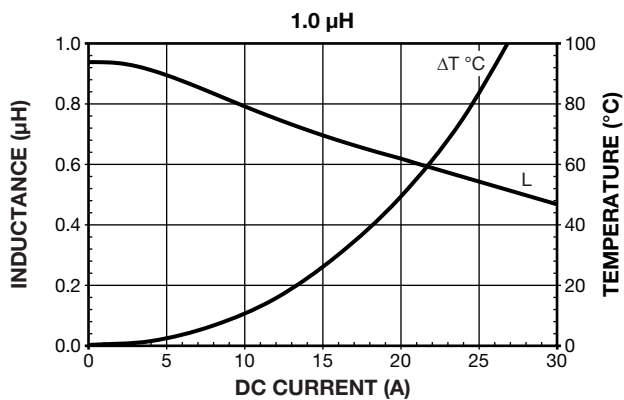
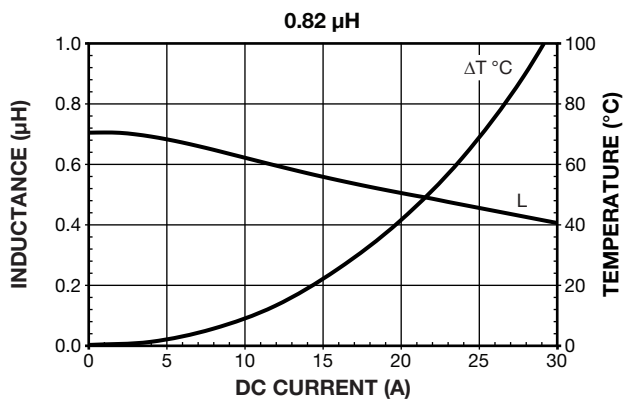
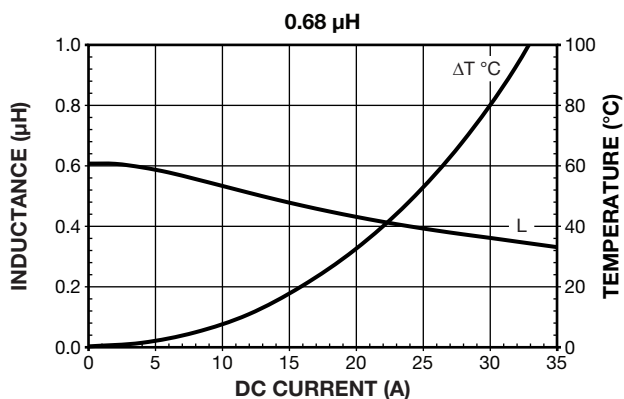
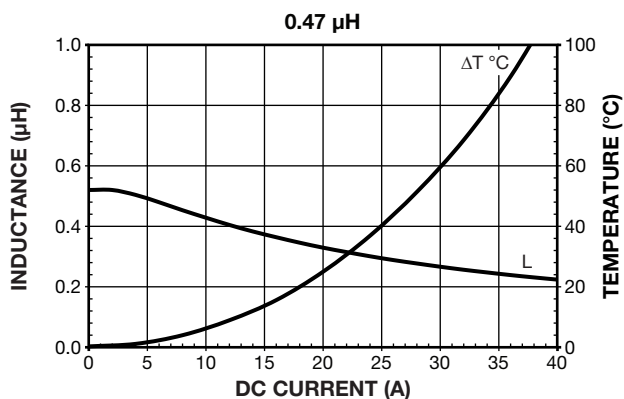
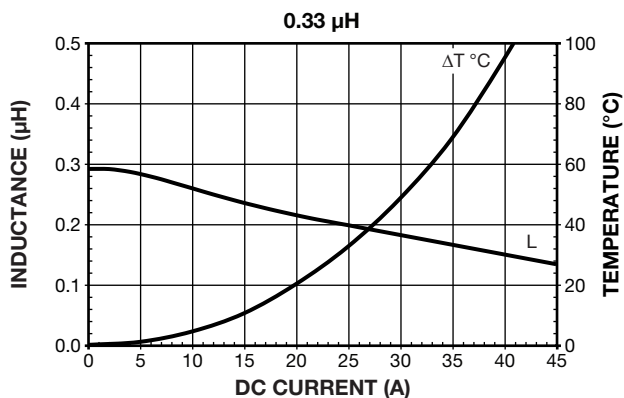
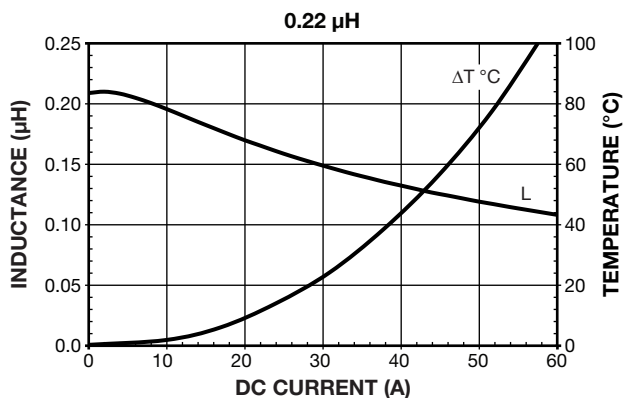
ER = tape and reel packaging (500 pcs on 13-inch reel)

Note

- For additional packaging details see "[Packaging Methods](#)"

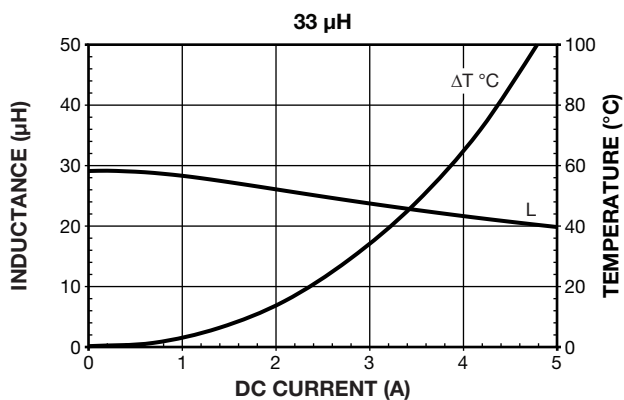
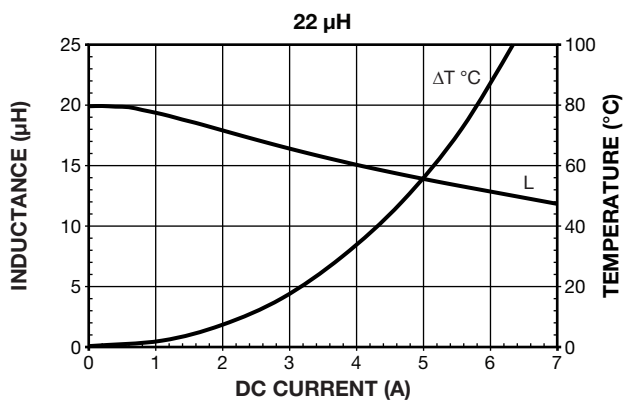
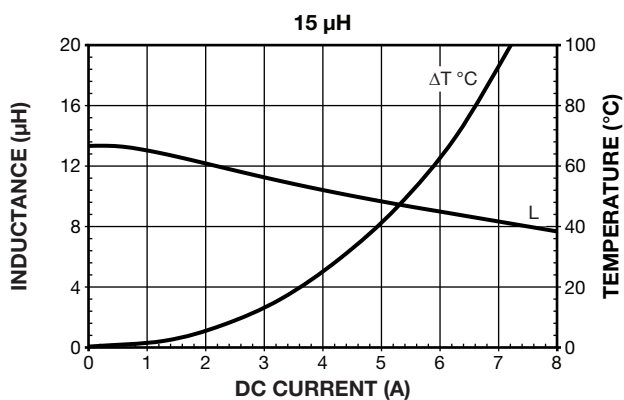
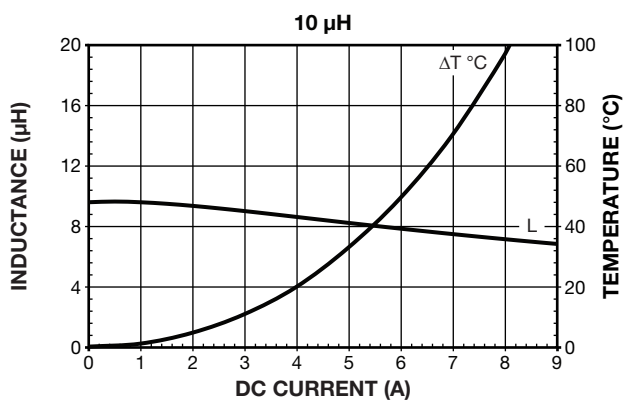
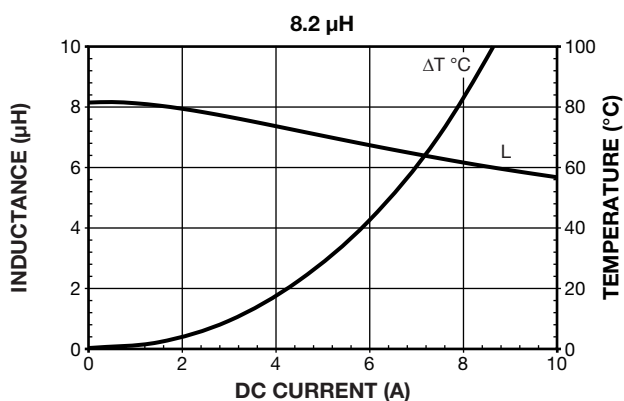
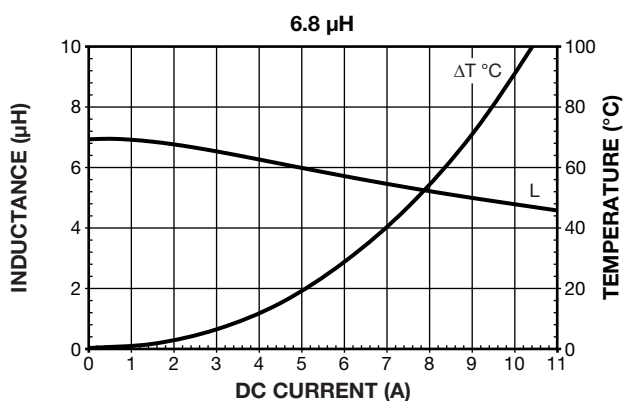
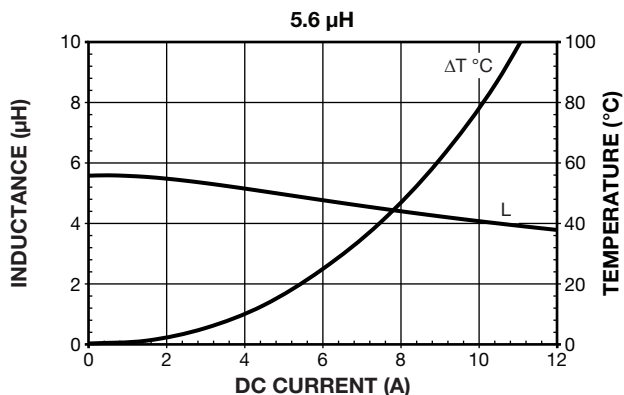
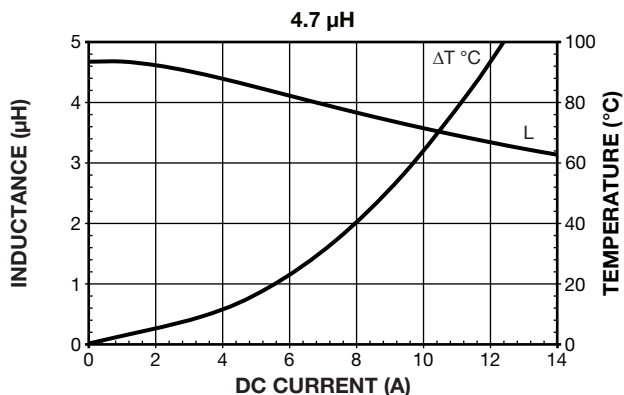


PERFORMANCE GRAPHS



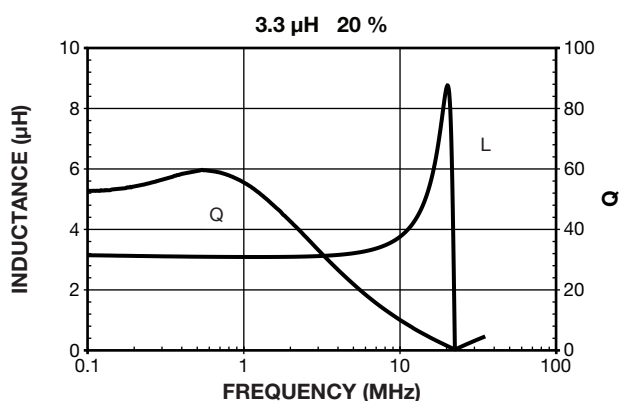
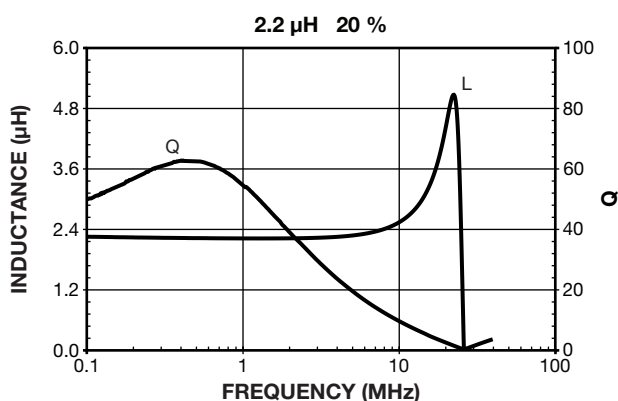
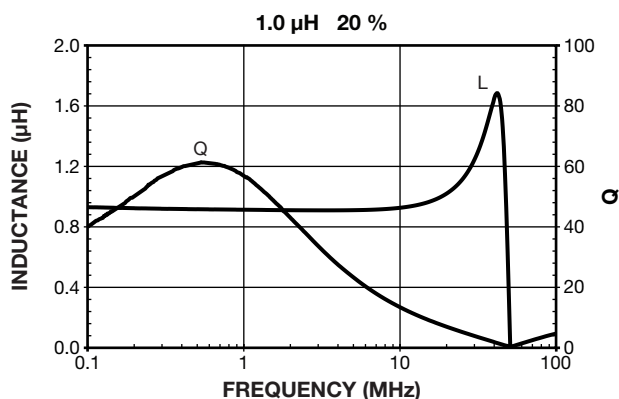
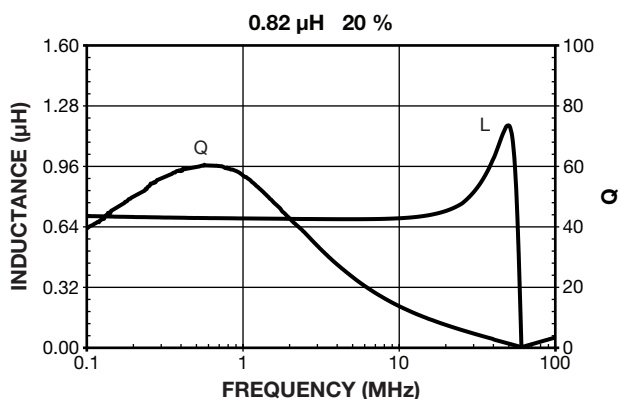
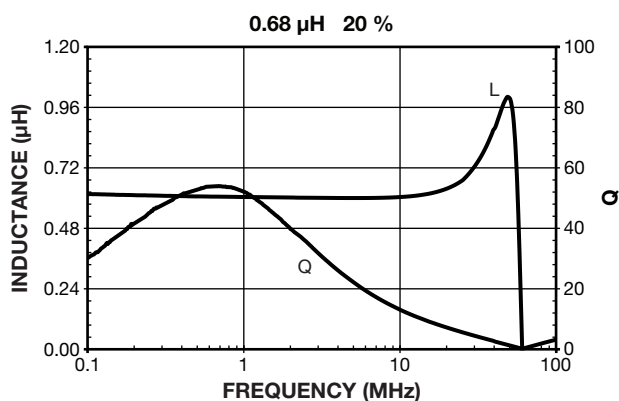
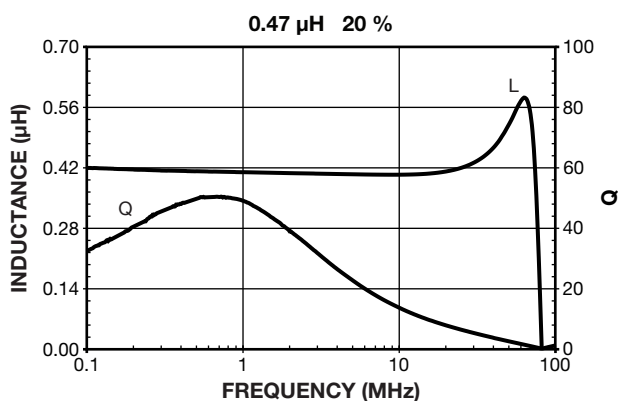
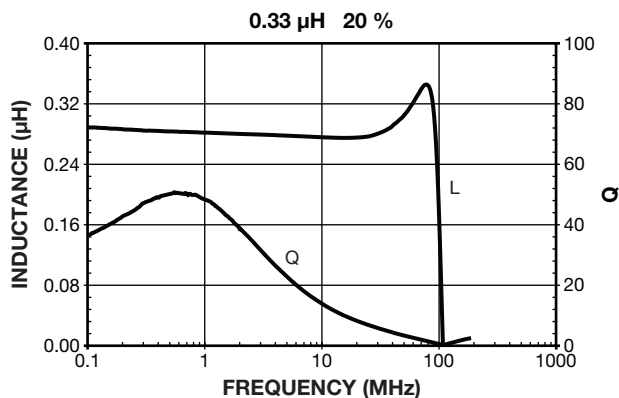
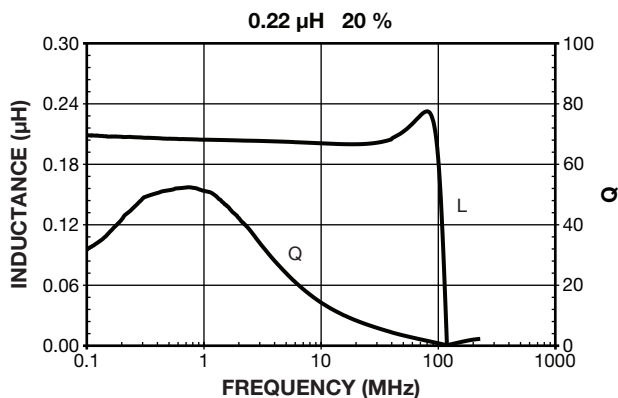


PERFORMANCE GRAPHS



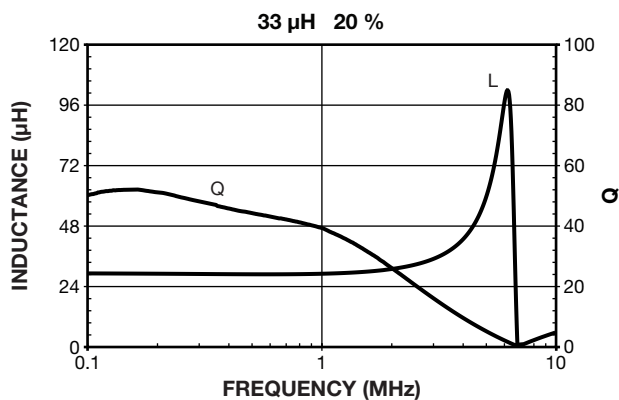
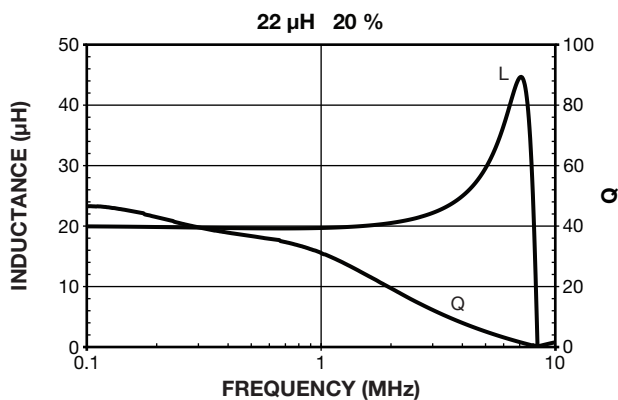
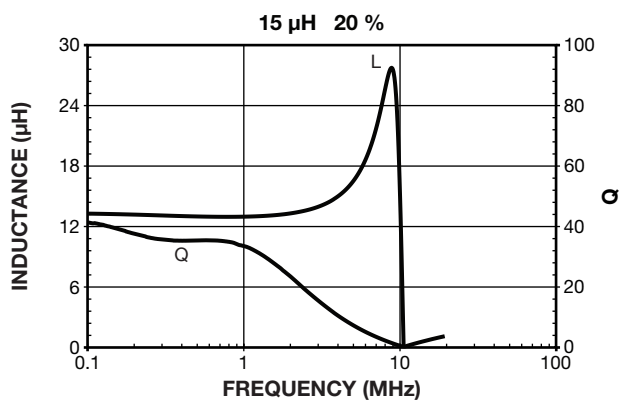
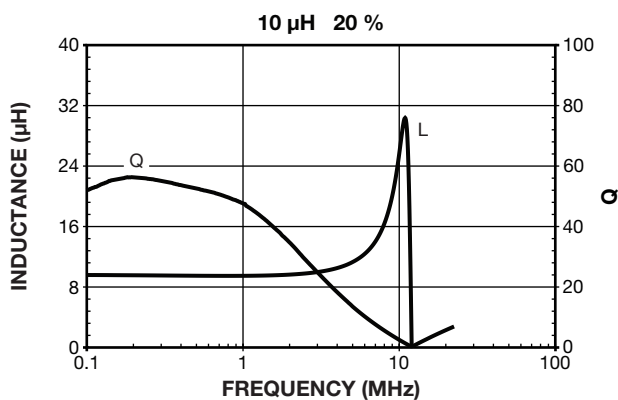
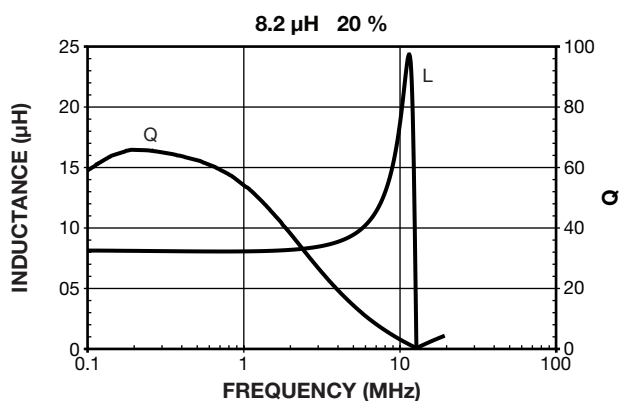
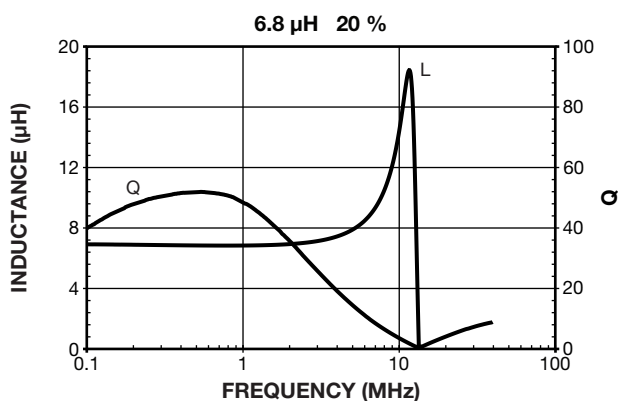
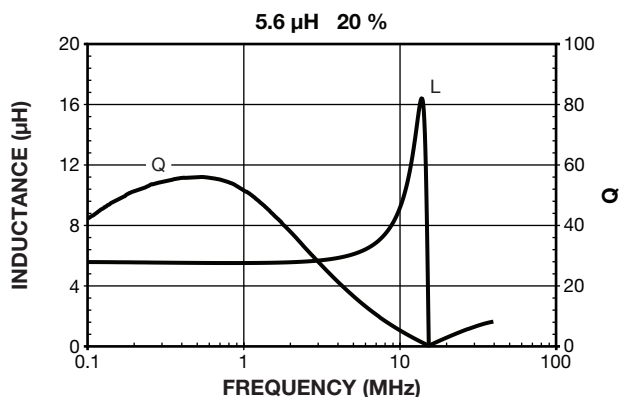
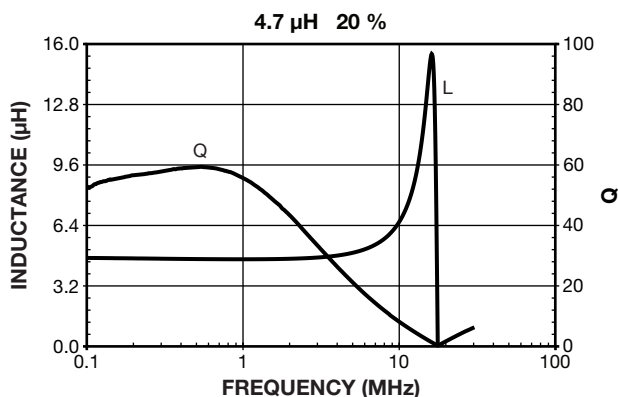


PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY





PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY





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